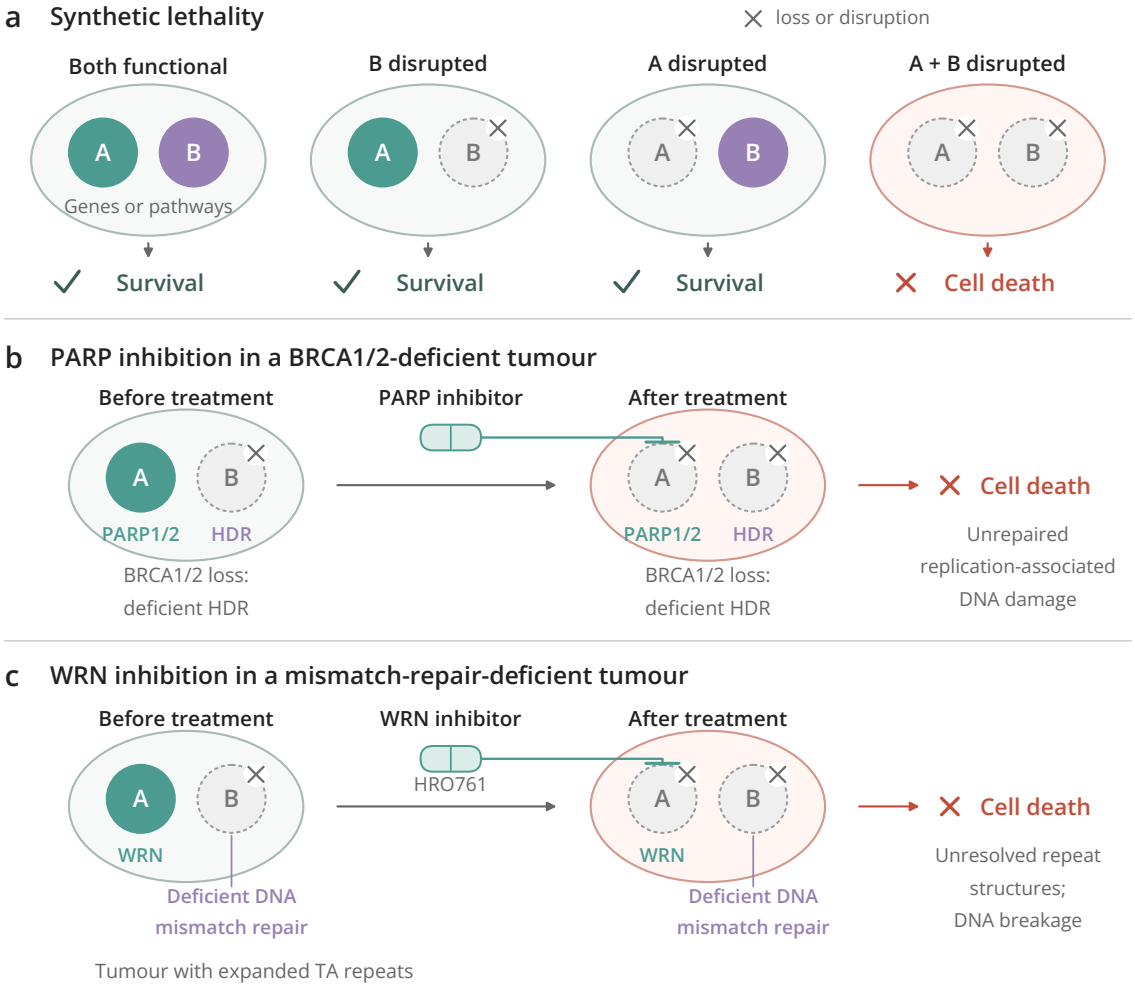
a
Synthetic lethality
loss or disruption
Both functional
B disrupted
A disrupted
A + B disrupted
A
B
Genes or pathways
Survival
Survival
Survival
Cell death
b
PARP inhibition in a BRCA1/2-deficient tumour
Before treatment
PARP inhibitor
After treatment
PARP1/2
HDR
BRCA1/2 loss:
deficient HDR
Cell death
Unrepaired
replication-associated
DNA damage
c
WRN inhibition in a mismatch-repair-deficient tumour
Before treatment
WRN inhibitor
After treatment
HRO761
WRN
Deficient DNA
mismatch repair
Cell death
Unresolved repeat
structures;
DNA breakage
Tumour with expanded TA repeats